

1. Company Name

ABC Startup Solutions

1.2 Nature of Business

ABC Startup Solutions is a software development company that provides web development, mobile application development, IT consulting, and digital solutions for small and medium-sized businesses.

1.3 Vision

To become a trusted technology company that provides innovative, reliable, and accessible digital solutions to businesses.

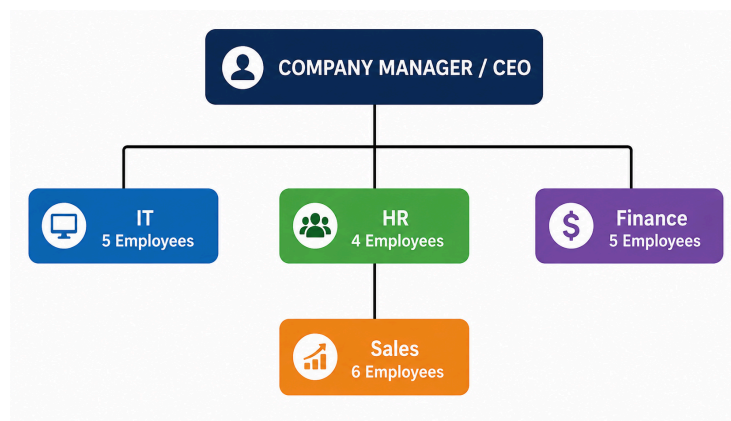
1.4 Mission

To develop quality software solutions that help businesses improve their operations, productivity, and digital services through modern technology.

1.5 Office Location

Santa Cruz, Laguna, Philippines

1.6 Organizational Structure



1.7 Employee Distribution

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
Total	20

2. ENTERPRISE HARDWARE INVENTORY

The proposed hardware inventory is based on the company's 20 employees and operational requirements. Desktop computers are assigned to regular workstations, while laptops are provided for employees who require mobility. A dedicated server, NAS storage, and backup drives are included to support centralized services and data protection.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	15	IT, HR, Finance, Sales	Daily office work and software development
HW-002	Laptop	5	IT / Management	Mobility, meetings, and remote work
HW-003	Server	1	IT	Centralized services and file storage
HW-004	Router	1	IT	Network and internet connectivity
HW-005	Network Switch	1	IT	Connect wired network devices
HW-006	Printer	2	HR / Finance	Printing documents and reports
HW-007	UPS	6	IT / Departments	Power protection
HW-008	Wireless Access Point	2	Office	Wireless network connectivity
HW-009	NAS Storage	1	IT	Centralized storage and backup
HW-010	External Backup Drive	2	IT	Offline data backup
HW-011	Monitor	20	All Departments	Workstation display

3. ENTERPRISE SOFTWARE INVENTORY

The proposed software environment provides the company with operating systems, productivity applications, development tools, security software, and system administration utilities.

Software	Version	License	Purpose
Windows 11 Pro	24H2	Commercial	Operating system for employee computers
Ubuntu Server	24.04 LTS	Open Source	Server operating system
Microsoft Office	Microsoft 365	Subscription	Documents, spreadsheets, and presentations

Visual Studio Code	Latest Stable	Free	Software development and coding
Git	Latest Stable	Open Source	Version control
GitHub Desktop	Latest Stable	Free	Git and GitHub repository management
VirtualBox	7.x	Open Source	Virtual machine management
Google Chrome	Latest Stable	Free	Web browsing and web applications
Microsoft Defender	Built-in	Included	Malware and threat protection
AnyDesk	Latest Stable	Commercial/Free Options	Remote technical support
7-Zip	Latest Stable	Open Source	File compression and extraction

3.1 Software Selection Justification

Windows 11 Pro provides the primary operating environment for employee computers. Ubuntu Server provides a suitable server operating system. Microsoft Office supports business productivity, while Visual Studio Code, Git, and GitHub Desktop support software development and version control. VirtualBox can be used for testing virtual machines. Google Chrome supports web-based applications and browsing. Microsoft Defender provides built-in security protection, AnyDesk supports remote technical assistance, and 7-Zip provides file compression and extraction.

4. ENTERPRISE NETWORK INVENTORY

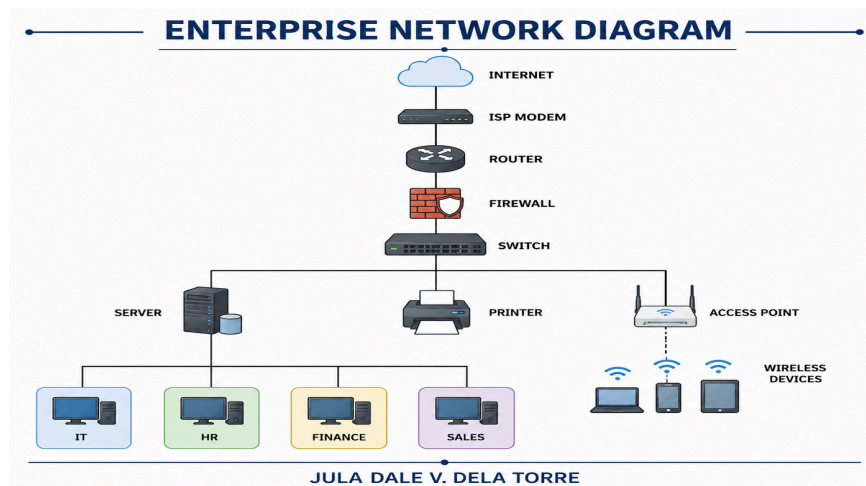
The proposed network infrastructure is designed to provide reliable connectivity and basic security throughout the company's single-office environment.

Asset ID	Equipment	Quantity	Purpose
NET-001	ISP Modem	1	Provides internet connection
NET-002	Router	1	Routes network traffic
NET-003	Firewall	1	Protects the network from unauthorized access
NET-004	Gigabit Switch	1	Connects wired devices
NET-005	Wireless Access Point	2	Provides wireless connectivity
NET-006	Patch Panel	1	Organizes network cable connections
NET-007	CAT6 Ethernet Cable	30	Connects network devices
NET-008	RJ45 Connectors	50	Terminates Ethernet cables

4.1 Network Design Explanation

The proposed network starts with the Internet connection provided by the ISP. The ISP modem connects to the router, which manages network traffic. The firewall provides an additional layer of security by controlling incoming and outgoing traffic. The firewall connects to the main switch, which distributes network connectivity to the server, printer, wireless access points, and departmental workstations.

5. ENTERPRISE NETWORK DIAGRAM



6. SYSTEM ADMINISTRATION ROLES

6.1 Helpdesk Technician

Responsibilities

- Troubleshoot hardware and software problems.
- Assist employees with technical concerns.
- Install and configure basic software.
- Handle support requests.
- Escalate complex technical issues.

Skills

- Troubleshooting
- Communication
- Customer service
- Basic networking
- Hardware and software knowledge

Common Tools

- AnyDesk
- Windows troubleshooting tools
- Ticketing systems
- Remote support tools

Certifications

- CompTIA A+
- ITIL Foundation

6.2 Network Administrator

Responsibilities

- Configure and maintain network devices.
- Monitor network performance.
- Manage routers and switches.
- Troubleshoot connectivity issues.
- Maintain network security.

Skills

- Networking
- IP addressing
- Routing
- Switching
- Network security

Common Tools

- Cisco Packet Tracer
- Wireshark
- Ping
- Command Prompt
- Network monitoring tools

Certifications

- CompTIA Network+
- Cisco CCNA

6.3 Linux System Administrator

Responsibilities

- Manage Linux servers.
- Install and configure server applications.
- Manage users and permissions.
- Monitor server performance.
- Perform server maintenance and backups.

Skills

- Linux administration
- Command-line skills
- Shell scripting
- Networking
- Server security

Common Tools

- Ubuntu Server
- Bash

- SSH
- systemctl
- Linux Terminal

Certifications

- Linux+
- LPIC-1
- Red Hat Certified System Administrator

6.4 Cloud Administrator

Responsibilities

- Manage cloud infrastructure.
- Configure cloud services.
- Monitor cloud resources.
- Manage cloud security and access.
- Maintain cloud backups.

Skills

- Cloud computing
- Networking
- Security
- Virtualization
- Infrastructure management

Common Tools

- AWS Console
- Microsoft Azure
- AWS CLI
- Cloud monitoring tools

Certifications

- AWS Certified Cloud Practitioner
- AWS Certified Solutions Architect
- Microsoft Azure Fundamentals

6.5 Collaboration of the Four Professionals

The four professionals work together to maintain the organization's IT infrastructure. The Helpdesk Technician handles employee technical concerns and resolves basic hardware and software problems. When an issue involves network connectivity, the Network Administrator investigates and resolves the problem. The Linux System Administrator manages Linux servers and ensures that server services remain available and secure. Meanwhile, the Cloud Administrator manages cloud-based resources and services. Through collaboration, these professionals help ensure that the company's systems, network, servers, and cloud services remain reliable and available.

7. INFRASTRUCTURE RECOMMENDATIONS

7.1 Internet Provider

A reliable business-grade fiber internet connection with at least 300–500 Mbps is recommended. The company will use cloud services, online communication, software development tools, video meetings, and web-based applications.

7.2 Server Specifications

The recommended server should use an Intel Xeon or equivalent processor, 32 GB RAM, at least 2 TB storage, RAID configuration, Gigabit Ethernet, and Ubuntu Server. These specifications provide sufficient processing power, memory, and storage for centralized company services and files.

7.3 Backup Strategy

The company should implement a 3-2-1 backup strategy: three copies of important data, two different storage media, and one copy stored off-site. A possible setup is Main Server → NAS Storage → External Backup Drive / Cloud Backup.

7.4 Security Recommendations

The company should implement firewall protection, strong password policies, regular software updates, user access controls, network security, regular backups, and security monitoring.

7.5 Antivirus

Microsoft Defender is recommended for Windows computers because it provides built-in protection against malware and other security threats.

7.6 Password Policy

Employees should use passwords with at least 12 characters, combine letters, numbers, and symbols, avoid password sharing, avoid easily guessed information, and use Multi-Factor Authentication for important accounts.

7.7 Expansion Plan

The company should use scalable network equipment, expandable storage, additional network ports, and cloud services to support future growth.

8. PERSONAL REFLECTION

This project taught me that proper IT infrastructure planning is an important part of system administration. Before purchasing equipment, installing servers, configuring networks, or deploying cloud services, a System Administrator must first understand the organization's requirements. Planning helps ensure that the selected technology is appropriate for the company's operations and future needs.

One of the things I learned from this project was how to identify the hardware, software, and networking requirements of a small business. I learned that choosing IT equipment is not simply about selecting devices; each item should have a clear purpose and should support the organization's daily activities. Preparing the hardware, software, and network inventories

helped me understand how different resources work together in an actual business environment.

The most challenging task for me was designing the enterprise network diagram. I had to consider how the Internet, ISP modem, router, firewall, switch, server, printer, wireless access points, and different departments should be connected. This task helped me understand the logical flow of a business network and the importance of organizing network components properly.

This project also showed me why planning is important before deployment. Without proper planning, an organization may purchase unnecessary equipment, experience connectivity problems, lack sufficient storage, or have security weaknesses. A well-planned infrastructure can reduce these risks and make future maintenance easier.

Overall, this project helped improve my skills in infrastructure planning, networking, technical documentation, and problem-solving. It also gave me a better understanding of the responsibilities of a System Administrator. In the future, I can use the knowledge gained from this activity when designing, implementing, maintaining, and troubleshooting IT infrastructure for an organization.

9. REFERENCES

- Laguna State Polytechnic University. ITEP 414 – System Administration and Maintenance: Week 2 Portfolio Project – Enterprise Infrastructure Planning for a Startup Company.
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